

**The Hidden Costs of Transferring: A Comparative Analysis of Outcomes for
Community College Transfer Students vs. Native Four-Year Students**

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I. Introduction

Community colleges are often touted as equalizers in the landscape of higher education. They provide accessible and affordable pathways for students from diverse backgrounds to attain an Associate's Degree. These institutions are particularly crucial for students without the financial means to directly enroll in a four-year university after high school and pay the costs of tuition. Among students who choose to enroll in community college, an overwhelming 81.4% aim to transfer to a four-year institution and earn a bachelor's degree (Handel and Williams 2012). In the case that these students are able to both successfully transfer and attain a bachelor's degree, they should in theory experience the same outcomes as "native" four-year students, while essentially paying half the price for their education. Despite the role of community colleges as facilitators of social and economic mobility, there are existing disparities between the aspirations of these students and their actual outcomes. Only 11.6% of community college students in the United States manage to attain a baccalaureate degree within six years (Handel and Williams 2012).

This paper seeks to address the question: Do students who transfer from a 2-year community college to a 4-year university achieve similar success as native 4-year students, while incurring lower direct costs? In this paper, success will refer to both the attainment of a baccalaureate degree and post-graduation earnings. I will posit that even after controlling for selection bias, there is an earnings penalty for transfer students due to late entry in the workforce. This delay often stems from the additional time required to complete a bachelor's degree at a four-year university caused by loss of credits during the point of transfer. Contributing factors to this credit loss include inadequate institutional advising at community colleges, strict credit

transfer policies at four year institutions, and disparities in social capital among transfer seeking students.

While the community college to four year transfer pathway offers clear advantages, such as incurring lower direct costs and similar signaling effects upon degree completion, these benefits are often overshadowed by significant barriers. Specifically, I will discuss a gap in bachelor's degree attainment, loss of credits at the point of transfer, and the risk of unsuccessful transfer. By examining empirical evidence from various studies, including "The community college pathway: An analysis of the costs associated with enrolling initially at a community college before transferring to a 4-year institution" by Hu, Ortagus, and Kramer, and "Are community college transfer students 'a good bet' for 4-year admissions? Comparing academic and labor-market outcomes between transfer and native 4-year college students" by Xu, Jaggars, Fletcher, and Fink, this paper aims to provide a comprehensive analysis of these barriers and their implications on the economic success of community college transfer students.

II. Barriers for Transfer Students

This section will examine the common obstacles faced by community college transfer students, focusing on the gap in bachelor's degree attainment, the loss of credits at the point of transfer, and the risk of unsuccessful transfers. One of the most prominent challenges is the gap in bachelor's degree attainment. However, there is contrasting evidence on whether this gap exists after controlling for selection bias. In the context of bachelor's degree attainment, selection bias refers to the distortion that occurs when the characteristics influencing a student's likelihood to transfer from a community college to a four-year university, such as academic ability or socioeconomic status, also impact their probability of successfully completing a bachelor's

degree, thereby skewing comparisons between transfer students and native four-year students. Handel and Williams, despite finding that “a student who starts at a community college is less likely to earn a baccalaureate degree in six years”, acknowledge that other researchers have found opposite results, which is why more research is needed to say this definitively. Empirical findings from Hu et al. show that “enrolling initially at a community college is negatively associated with a 37.4% decrease in the likelihood of obtaining a baccalaureate degree”. Simultaneously, Xu et al. state that community college transfers “had a comparable probability of baccalaureate attainment compared with native 4-year students with a similar background and academic record”. Given the room for debate on this topic, I will first examine factors which may contribute to this observed gap in degree attainment.

Community college transfer students face several unique challenges that contribute to their lower rates of bachelor’s degree attainment compared to their peers who begin at four-year institutions. In an article written by the CollegeBoard, Handel and Williams identify one of these factors as the disparity between curricula at community colleges and four-year institutions. Community colleges typically offer only lower-division courses, which may not align with the upper-division courses required at four-year schools. Handel and Williams explain that at four-year institutions, “the lower-division preparatory courses...align with the upper-division major curriculum because [the] faculty are in close contact with one another, and there are layers of oversight.” This misalignment can leave transfer students underprepared, making it difficult for them to succeed in advanced courses.

Additionally, the academic environment at community college often lacks opportunities supporting extracurricular and campus engagement that are prevalent at four-year schools, which are meant to support academic growth. At these institutions, “academic engagement is supported

and encouraged not only in the classroom but also in the activities outside of class that supplement and extend learning” (Handel and Williams 2012). Thus it has been argued that this difference in curriculum has contributed to a “transfer shock” among community college transfers, referring to “a decline in transfer students’ GPA after the first term at the receiving institution” (Handel and Williams 2012). This highlights that even when comparing a transfer student to a native four-year student with similar abilities, we can still expect differences in academic performance due to the adjustments transfer students must make to align with the new curriculum. This shock may contribute to a higher dropout rate among community college transfers.

Another barrier faced by transfer students is credit loss at the point of transfer. This can lead to increased time in school, which also means more semesters of tuition payments beyond the originally anticipated two years. Given that the community college transfer path is promoted as being an affordable path, the potential to spend more semesters at a four year university becomes extremely problematic for this argument. As Handel and Williams note, “students who are preparing for transfer are often faced with insufficient information in print or on the Web about the transfer process [and] nonexistent or indecipherable policies specifying how their community college credit will transfer.” This lack of clarity can contribute to the loss of credits when students attempt to transfer. This finding is supported by Xu et al., who discovered that vertical transfer students earned 18 more credits than necessary for a traditional four-year degree, suggesting either that they lost credits during the transfer process or were required to take additional major-specific courses at the four-year institution. As a result, transfer students often spend more terms in college than their native four-year counterparts and are more likely to still be enrolled years after their initial college enrollment. Late entry into the workplace can

potentially impact transfer students' earnings due to the forgone time spent finishing their degree. Delayed entry into the labor market has also been found to be negatively associated with job placement (Hu et al. 2018). Thus transfer students face multiple risks that not only prolong their time in school, but also delay entry into the workforce, ultimately affecting their earnings potential.

One of the most obvious risks associated with the transfer route is the possibility of not being admitted to a four-year university due to the selectivity of these institutions. National estimates suggest that only 25 to 35 percent of community college students successfully transfer to a four-year institution, highlighting the significant barriers to transfer success (Handel and Williams 2012). This low success rate underscores the large proportion of students who are derailed from their plans to transfer. This can be both due to selectivity of four-year universities as well as the complexity of the transfer process. In a Quantitative Comparative Analysis (QCA), Jabbar et al. discuss how even students who receive strong advising and feel a sense of belonging can be hindered in their transfer plans if they lack financial support or necessary information. For instance, "significant out-of-school obligations, intimidation about applying to college, and lack of financial knowledge about funding higher education could lead to a failure to transfer" (Jabbar et al. 2019). These gaps – whether financial, informational, or related to advising – are all significant factors in preventing successful transfer processes. While these challenges are not directly addressed in the empirical studies by Xu et al. (2018) or Hu et al. (2018), as their focus is mainly on comparing successful transfer students to native four-year students, it is important to acknowledge that the risk of not being able to transfer is a legitimate and significant barrier for many community college students.

III. Analysis of Literature and Discussion

In the academic article titled “The Community College Pathway: An Analysis of the Costs associated with Enrolling Initially at a Community College Before Transferring to a 4-year Institution” by Xiaodan Hu, Justin Ortagus, and Dennis Kramer, the authors seek to identify the influence of initial community college enrollment on three main outcomes: bachelor degree attainment, cumulative student loan debt, and the time-to-degree. Their results indicate that students who first enrolled in community college before transferring were less likely to earn a bachelor’s degree, accumulated less student loan debt, and took approximately one semester longer to graduate compared to their peers who began their studies at a 4-year institution. I will compare Hu et al.’s empirical methods to those of Di Xu, Shanna Jaggars, Jeffrey Fletcher, and John Fink, in the paper titled “Are Community College Transfer Students ‘a Good Bet’ for 4-Year Admissions? Comparing Academic and Labor-Market Outcomes Between Transfer and Native 4-Year College Students”, which seeks to answer the question “Is a given student at a 4-year college who has a given GPA and credit accumulation at the beginning of a particular term more likely to be successful if he has just arrived from a community college or if he has earned that same record while at the 4-year institution?” Both papers use matching techniques in order to account for selection bias, and use similar statistical methods such as logistic and linear regression. However, Xu et al. found that there was not a significant difference in the probability of a transfer student completing a bachelor's degree after employing a matching technique which matched students within the same institution. They also found that vertical transfer students had a significantly longer time-to-degree which may have influenced quarterly earnings 8 years after college entry. Given the complexity of the empirical methods utilized in each paper, it is worth

comparing their matching techniques to analyze the ability of each paper to effectively mitigate selection bias.

Hu et al. use propensity score weighting (PSW) to account for selection bias, comparing outcomes from students who started at community colleges (treatment group) to those who began at 4-year institutions (control group). Propensity score weighting is often used in observational studies where treatment assignment is not random. This method helps to balance predictive variables between groups to make more accurate conclusions. They use data from the Beginning Postsecondary Students Longitudinal Study (BPS: 04/09), a nationally representative dataset. Their final sample included 2,410 treated students and 4,640 control students, with outcomes of interest being (1) baccalaureate degree attainment, (2) cumulative loan debt, and (3) time-to-degree. In order to measure these outcomes, they apply a logistic regression to estimate the effect of treatment on baccalaureate degree attainment. Logistic regression is used for predictions of binary outcomes, which is either graduating from the four-year school or dropping out of the four-year school. To estimate the effect of the treatment on cumulative loan debt and time to degree, an ordinary least squares (OLS) regression is implemented. Their generic regression model takes the following form:

$$\gamma_i = \alpha_{1i} + \beta_1 Treat_i + \beta_2 X_{1i} + \varepsilon_{1i}$$

In this model, the response variable represents each of the three outcomes, while $Treat_i$ is a “dummy variable” that will take on the value “1” if the students received the treatment (transferring from a community college) and takes on the value “0” if they began at the four year institution. X is a vector of student level covariates, or independent variables that can influence a given statistical trial, which includes demographics, pre-college characteristics, and family background.

The PSW approach involves two main steps. First, Hu et al. estimate the probability of enrolling initially in a community college (the propensity score) based on the observed covariates. This allows them to estimate the average treatment effect on the treated, or the ATT.

$$ATT \equiv E(y_1 - y_0 \mid W=1)$$

ATT can also be understood as the difference in proportions of subjects experiencing the event in the two groups. In this equation, y is the outcome of individual i if they are subject to the treatment, while W is the treatment indicator. Then, they use these scores to assign weights so that the weighted control group resembles the treated group in terms of observed characteristics.

$$ATT - W = Treatment + (1 - Treatment) \cdot \left(\frac{p}{1 - p} \right)$$

Here Treatment is the same dummy variable, while p is the probability of receiving the treatment. Thus when $W=1$, the PSW model looks at the outcome for those enrolled first in a community college. When $W=0$, the weight on the outcome of the native four-year is equal to p . Therefore higher propensity scores in control units ($W=0$) will generate larger ATT weights.

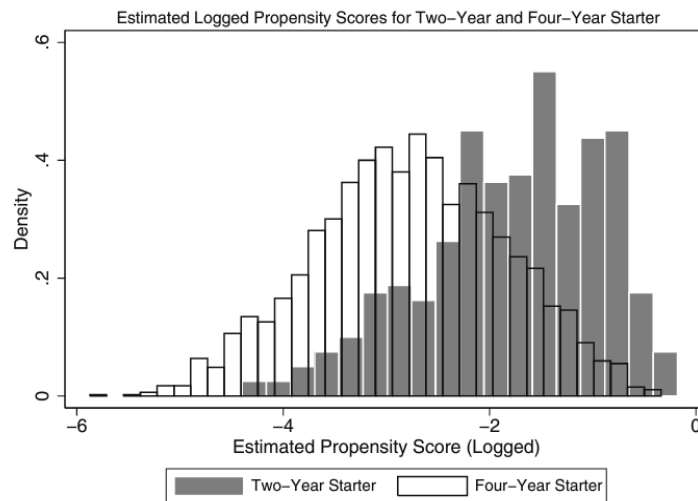


Figure 2. Overlap in estimated propensity scores between the treatment and control groups in the completer-only sample.

Figure 2 visualizes the overlap in propensity scores of community college transfer students and native four-year students. The distribution of native students is skewed to the left, indicating that their probabilities of enrolling in a community college were much lower. This is presumably because of selection bias, where the control group is more likely to directly enroll in a four-year college due to variables such as ability. Those with the highest ability (test scores, GPA, etc.) are more likely to be accepted into a four-year college, which are much more selective with who they admit than community colleges.

Hu et al. first present their results for the logistic regression, measured in likelihood of bachelor degree completion¹. They find that initial community college enrollment is associated with a 37.4% decrease in likelihood of attaining the degree, even after accounting for pre-college differences using the propensity score weights. This was denoted as a statistically significant difference with a p value of $p < 0.001$, suggesting strong evidence against a different mean likelihood of baccalaureate degree completion between the two samples. The linear regression model of cumulative loan debt and time-to-degree on the treatment and pre-college covariates only included bachelor degree completers. For cumulative loan debt, the weighted sample estimates transfer students on average borrowed \$1700 less than four-year natives. This result did not come back as statistically significant, which may indicate that there is not a consistent pattern of cumulative loan debt between the two groups. The last examined result was an estimated difference in time-to-degree, which decreased from 5.1 months in the unweighted model to 2.4 months in the weighted model. The first result was denoted as statistically significant with $p < 0.05$. However, I believe it is more meaningful to also interpret the effect size in this case rather than the p value alone. An increase in time-to-degree of 2.4 months

¹ Discussed results from Hu et al. will be included in Appendix A.

corresponds to about an extra semester spent in school. This may be considered a significant barrier depending on the goals and socioeconomic background of a given transfer student.

Although Hu et al.'s propensity score weights are a matching measure taken to create a somewhat “synthetic” control group, there are limitations to this method that I believe make the study less comparable to a randomized sample where we might assign similar students to either community college or a four-year university. While this method reduces selection bias by balancing characteristics in the treatment and control group, its effectiveness is contingent on accurately modeling the propensity score. If any key covariates are omitted, such as student motivation or institutional quality, then corresponding results are only as accurate as the propensity score model. PSW assigns weights based on the probability of treatment but does not inherently address these potential unobserved differences which could correlate with both treatment assignment and outcomes. Additionally, Hu et al. match students across all institutions, potentially neglecting variation in institutional quality. For example, a transfer student may lose credits or face structural hurdles at specific universities, which might not be captured in their model. This mismatch could lead to imprecise estimations of the true causal effect.

By comparison, Xu et al. use multiple matching strategies to refine their analysis. Their data come from the State Council of Higher Education for Virginia (SCHEV) where about 10% of the individuals out of 42,803 students were “vertical” community college transfer students. The two primary outcomes measured were (1) probability of attaining a bachelor's degree and (2) labor market earnings eight years prior to college entry. Once again a logistic regression is used to estimate the binary outcome:

$$Probit(BAattainment_i) = \alpha + \beta Twoyear_i + X_i + \mu_i$$

In the logistic regression, “*Twoyear*” is a dummy variable (similar to “*Treat*” in Hu et al.’s model) where its coefficient Beta equals 1 if the student enters the four-year school as a community college transfer. X is also a vector of controls such as age, race, financial aid status, and ESL status. To estimate labor market earnings Xu et al. use a linear regression:

$$Y_i = \alpha + \beta Twoyear_i + Exp_i + Exp_i^2 + X_i + \mu_i$$

Y is measured as the average quarterly earnings of individual i in 2012 and the Exp term represents the student’s labor market experience in 2012, with a quadratic term included for the possibility of a nonlinear relationship between work experience and earnings (based on prior literature). There were four ways that Xu et al. ran each of these regressions, corresponding to three different matching techniques. The first method did not match students, restricting the sample to students with a baseline amount of credits earned before transferring, using a 48-credit restriction. The second maintained the same credit constraint, while also utilizing a cross-institutional propensity score to match students similar in terms of demographics. During this process, there was such a large population of native 4-year students compared to vertical transfers that almost every vertical transfer student was matched within 0.1 standard deviations of their propensity score. The third method also used the propensity score matching by demographics, but rather than using credit restrictions matched based on college level credits and prior GPA. This led to a larger sample size and a more representative sample of community college transfer students. In the fourth matching method, Xu et al. maintain the same criteria in the third, but add an institution-specific criteria to match students attending the same four year institution.

Xu et al.'s non-matched results² shows that vertical transfer students were 17% less likely to attain the bachelor's degree than the native four year students, a highly significant difference at $p < 0.01$. This result changed after matching across institutions (Strategy 3), showing that transfer students had a 4% higher probability of completing a bachelor's degree compared to native students, which not only contradicts the finding of Hu et al.'s propensity weighting model, but makes the opposite conclusion. This came back as statistically significant with $p < 0.01$. However, after matching within institutions (Strategy 4), this difference was no longer statistically significant, with a 2% higher likelihood of graduating from the four-year with $p > 0.1$.

There was a significant difference between labor market earnings between the two groups in both the third and fourth matching strategies, with $p < 0.01$ in both regressions. Across institutions, the average quarterly earnings of vertical transfer students in 2012 were \$898 lower than that of the control group, and this difference increased to \$947 when matching within institutions. Suspecting that this difference in earnings is attributed to late entry into the labor market, Xu et al. use their fourth matching strategy to compare the number of credits earned by 2-year entrants versus 4-year entrants. They found that the number of credits for baccalaureate earners differed by 18 credits, with vertical transfer students averaging 139 credits compared to 121 credits in the native four-year group. This effect size was highly statistically significant, with a test statistic of 21.65 (standard error = 0.85). Xu et al. acknowledge that they could not track the credits lost at the point of transfer, but posit that this difference is due to failure to transfer credits over to the four-year or requirements to earn additional major specific credits. This result corresponded with a 3 semester difference in enrollment, with transfers on average being enrolled for 15 semesters compared to 12 for the control group.

² Discussed results from Xu et al. shown in Appendix B.

While both empirical studies employ rigorous matching techniques to control for selection and ability bias, I believe Xu et al. 's fourth matching strategy provides the closest approximation to a randomized controlled experiment for several reasons. By matching students based on college GPA and accumulated credits at the time of transfer, Xu et al. account for important measures of the students' academic progress, which limits the effects of ability bias. This directly compares students with their most updated academic records, unlike Hu et al. who rely on high school GPA, a pre-college measure which may not accurately reflect a student's trajectory in higher education. Pairing transfer students by number of credits obtained is also an effective control for selection bias, as number of credits earned may be an indication of motivation or ambition among the students. Furthermore, Xu et al.'s results provide a closer estimate to the true effect of transfer outcomes by implementing institutional comparability. Matching within the same institution controls for unobservable differences in institutional quality, academic policies, and credit transfer agreements. These are critical factors to make generalizable statements about differences in outcomes between the two groups, but are not addressed in a cross-institutional analysis like that of Hu et al.

The findings from both studies have important implications for understanding the cost effectiveness of the community college transfer path. Hu et al.'s work highlights that at a national level, students transferring from community colleges face significant barriers in baccalaureate attainment. The savings they experience in loan debt is modest, and seemingly insignificant especially after accounting for the increase in time-to-degree. By contrast, Xu et al. show that when institutional and academic factors are rigorously controlled for, transfer students can achieve academic outcomes comparable to their native 4-year peers. However, both studies underscore that there are critical economic challenges faced by transfer students. Xu et al.'s

estimate that transfer students require an additional three semesters to graduate reveals a significant financial burden. These extra semesters not only increase direct tuition costs but also delay entry into the workforce, thereby compounding the opportunity costs of extra time in school. Hu et al.'s findings on loan debt further support this implication; the debt accumulation for community college transfers does not differ enough from their four-year counterparts, meaning transfers do not maintain their cost advantage if they must spend additional semesters at a more expensive institution.

To fully evaluate the economic impact of community college transfer, we must analyze whether the foregone earnings from additional time spent in college exceed the tuition savings achieved by starting at a community college. Xu et al.'s findings suggest that transfer students incur substantial opportunity costs due to delayed graduation, particularly in the form of lost earnings during the additional three semesters they spend in college. For example, if the average entry-level salary for a bachelor's degree holder is \$50,000 annually, the additional time in school results in approximately \$12,500 of lost wages per semester. When combined with higher tuition costs at 4-year institutions for these semesters, the total economic burden may outweigh the initial tuition savings gained through community college enrollment. This emphasizes the need for policies that streamline the transfer process to minimize delays, such as creating uniform standards for credit transfers and reducing variability in institutional policies. By addressing these structural discrepancies, the higher education system in America can better support the success of transfer students.

IV. Conclusion

The research question driving this paper focused on whether students who transfer from two-year community colleges to four-year institutions achieve success comparable to native four-year students, while incurring a lower direct cost. Success was defined through two main components: attainment of a baccalaureate degree and post-graduate earnings. Community college transfer has long been recognized as an equitable path for high-achieving students without the means to afford four years of tuition at selective colleges. However, the findings of both Hu et al. and Xu et al. reveal a more complex picture of the tradeoffs associated with initial enrollment at community colleges. While transfer students seemingly can achieve similar academic outcomes at a four-year institution when controlling for ability and demographics, they often face systemic barriers that prolong their time-to-degree and delay entrance into the workforce. Depending on how drastic time-to-degree varies, this could potentially undermine the cost advantage of going to a community college and even impact long-term earnings. Therefore, students planning to attend a community college and transfer to a four-year school should be made aware of these barriers so they can adequately prepare for a more seamless transition between schools and obtain credits that will be accepted by universities and colleges of interest.

There are also strategies that institutions and policymakers themselves can implement to support these students. Strengthening communication and partnership between community colleges and four-year universities can minimize credit loss and streamline the degree pathway, as discussed by Handel and Williams. Developing state-wide or regional transfer systems with uniform policy could be a first step in simplifying the process for many students. Perhaps more importantly, providing comprehensive advising at the community college level is critical. As Jabbar et al. argue, lack of information and poor counseling are major reasons for failure to successfully transfer. It can be assessed that credit loss at the point of transfer is likely avoidable

when community college students have access to advisors that can align their coursework with requirements at other institutions. Lastly, encouraging a closer alignment of curriculum between community colleges and four-year colleges can ensure that lower-division courses prepare students for their upper-division requirements to reduce the risk of academic delay.

In the current educational landscape, evaluating the economic benefits of the transfer pathway will involve a careful consideration of whether foregone earnings from additional time spent in college outweigh tuition savings achieved by starting at community college. Addressing these barriers through improved credit transfer systems, enhanced advising, and curricular alignment could help to ensure that the community college pathway truly delivers on its promise of accessible and cost-effective higher education.

Sources

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V. Appendix A - Results of Hu et al. (2018)

Table 3 Descriptive results for degree completion (%)

	<i>Unweighted</i>			<i>Weighted</i>		
	<i>Treated (2-Year)</i>	<i>Control (4-Year)</i>	<i>Difference</i>	<i>Treated (2-Year)</i>	<i>Control (4-Year)</i>	<i>Difference</i>
Bachelor degree completion	0.290 (0.454)	0.595 (0.491)	−0.305	0.290 (0.454)	0.432 (0.495)	−0.142

Standard deviation in parentheses.

Table 4 Likelihood of bachelor degree completion

	<i>Logit</i>	<i>PSW</i>	<i>NN</i>	<i>Kernel</i>
First institution: 2-year	0.581*** (0.088)	0.626*** (0.069)	0.651*** (0.046)	0.712*** (0.016)
Number of observations	6630	6630	3360	6630
Weighted observations	No	Yes	No	No
Pre-college covariates	Yes	Yes	Yes	Yes

Note: Odds ratios presented; *logit* logistic regression, *PSW* propensity score weighting, *NN* nearest neighbor matching. Sample includes students starting a public 2-year or public 4-year institutions initially and received a bachelor degree within 6 years. Standard errors in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 5 Descriptive results for time-to-degree and cumulative loan debt

	<i>Unweighted</i>			<i>Weighted</i>		
	<i>Treated (2-Year)</i>	<i>Control (4-Year)</i>	<i>Difference</i>	<i>Treated (2-Year)</i>	<i>Control (4-Year)</i>	<i>Difference</i>
Time-to-degree (months)*	50.7	45.6	5.1	49.8	47.4	2.4
Cumulative loan debt (\$)						
Baccalaureate degree completers	14,530	12,986	1544	14,839	16,539	−1700
Pooled sample	12,518	12,780	−262	13,328	15,694	−2366

Note: TTD sample includes students starting a public 2-year or public 4-year institutions initially and received a baccalaureate degree within 6 years.

VI. Appendix B - Results of Xu et al. (2018)

Table 6. Baccalaureate attainment and labor-market outcomes of transfer students versus native 4-year college students.

Matching Strategy and Outcome	Estimate	Standard Error	R ²	N
1: Without matching (48-credit restriction)				
Baccalaureate attainment	-.17***	.01	.06	35,873
Average quarterly earnings, 2012	-	136	.06	18,954
	\$2,531***			
Average quarterly earnings, 2012 (conditional on bachelor's degree attainment)	-	169	.05	15,414
	\$2,481***			
2: Propensity score matching across destination colleges (48-credit restriction)				
Baccalaureate attainment	-.02*	.01	.07	6,017
Average quarterly earnings, 2012	-\$664	520	.07	3,923
Average quarterly earnings, 2012 (conditional on bachelor's degree attainment)	-\$388	761	.06	2,473
3: Propensity score matching across destination colleges (matched on credits prior to transfer)				
Baccalaureate attainment	.04***	.01	.11	6,695
Average quarterly earnings, 2012	-\$411***	150	.10	4,595
Average quarterly earnings, 2012 (conditional on bachelor's degree attainment)	-\$898***	202	.11	2,917
4: Propensity score matching by destination college (matched on credits prior to transfer)				
Baccalaureate attainment	.02	.01	.13	6,530
Average quarterly earnings, 2012	-\$891***	150	.11	4,514
Average quarterly earnings, 2012 (conditional on bachelor's degree attainment)	-\$947***	195	.12	2,864

* $p < .1$. *** $p < .01$.

Table 7. Enrollment outcomes for vertical-transfer students versus native 4-year college students.

Outcome	2-Year Entrants	4-Year Entrants	Effect Size
All matched students			
Number of credits (any type)	123	103	20.35 (1.01)***
Number of college-level credits	118	102	15.78 (1.00)***
Number of semesters enrolled	14	11	2.61 (.12)***
Still enrolled at beginning of 2012	38%	32%	0.06 (.01)***
Number of postexit quarters	4	6	2.41 (.01)***
Baccalaureate earners			
Number of credits (any type)	139	121	18.14 (.85)***
Number of college-level credits	133	120	13.72 (.84)***
Number of semesters enrolled	15	12	2.41 (.13)***
Number of postexit quarters	4	5	1.22 (.02)***

Note. The sample was constructed using Strategy 4 (within-institution match including credits accumulated at the time of transfer).

*** $p < .01$.